Yash Kishore Maheshwari

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(617) 777-8735 | Boston, MA | Seeking Jan–Aug 2027 Co-op (4–8 months) or Summer 2027 Internship

EDUCATION

Northeastern University, Boston, MA — Expected May 2029
B.S. in Computer Science, University Honors Program — GPA: **3.99/4.00** | Dean's List
Coursework: Algorithms, Computer Systems, Systems Security, Databases, Cloud Computing, Object-Oriented Design
Certifications: AWS Certified Solutions Architect – Associate, Google Cybersecurity Professional Certificate

EXPERIENCE

Undergraduate Research Assistant — Jul 2026 – Present
Northeastern Khoury Algorithms and Theory Group | *Chain-of-Thought Complexity* | *PEAK Award* — *Boston, MA*
- Implemented 10 transformer constructions in Python and NumPy from mathematical specifications, building a verification suite covering **1.7M+ test cases**, including exhaustive checks over selected input ranges.
- Audited stated assumptions against what the proofs use, reporting 11 defects, one prompting a new definition.
- Replaced one-hot positional encoding with spherical codes, cutting **2,004 dimensions to 37**.
- Backed every reported finding with a re-runnable test in the repo, a standard the faculty lead adopted team-wide.

Teaching Assistant, CS 2100 — Jan 2026 – Present
Northeastern University, Khoury College of Computer Sciences — *Boston, MA*
- Led 2 weekly labs and 5 office hours a week for **200+ students** on data structures, pandas, and graph algorithms, debugging live with them and covering Git, lint rules, and interview-style problems outside the syllabus.
- Evaluated homework and codewalks across 8 assignments with feedback on structure, test coverage, and design.
- Brought recurring misconceptions from office hours to course staff, changing how those topics were re-taught.

Data Visualization Lead — Sep 2025 – Present
Northeastern Electric Racing — *Boston, MA*
- Engineered an unattended post-race pipeline in Python, pandas, and Matplotlib that **cuts processing time ~40%** and renders a frame-synced telemetry overlay (speed, throttle, brake, pack temp) onto race video for driver debriefs.
- Published **nercardata** to PyPI after gathering requirements from 4 engineering subteams, standardizing sensor queries so new members could pull clean telemetry without learning the database schema.

PROJECTS

DepWatch: Vulnerability Intelligence Platform | *Python, Airflow, AWS, Gemini API* — Sep 2026 – Present
- Built the Airflow pipeline behind a vulnerability scanner, landing GHSA and OSV data in MinIO and transforming it into a **33,874-record** PostgreSQL corpus matched against a repo's dependency manifest.
- Added a Gemini API layer that assesses whether a flagged package is reachable in the calling code, cutting hundreds of version matches to the few worth acting on; it then writes each fix as a prompt a coding agent can execute.
- Indexed the corpus for search by description rather than CVE ID, using Qwen3 embeddings and an HNSW index in pgvector. Deployed on AWS ECS Fargate, RDS, and MWAA with Terraform.

ImpactTrail | *Next.js, TypeScript, Prisma, PostgreSQL, Gemini API* — Feb – May 2026
- **Best Overall Winner, HackBeanpot 2026**. Led the 5-person team and ran the pitch.
- Wrote the prompt layer that turns a resident's complaint into a volunteer quest, feeding the Gemini API every live quest and validating each response against a Zod schema built from the Prisma enums.
- Modeled volunteer-to-quest matching as a **min-cost flow** capped per volunteer and per quest, over a 13-table schema.
- Set up GitHub Actions CI with type-check, build, and automatic rollback on a failed deploy.

NU Rate-on | *React, Tailwind CSS, Puppeteer, Supabase* — Nov 2025
- Ranks campus dining halls head-to-head using student votes. Retrieved menu JSON from the Cloudflare-protected dining API through Puppeteer browser sessions, normalizing halls, meal periods, and per-item nutrients into five tables.
- Wrote the vote and menu APIs on Supabase, with get-or-create ingestion so a daily re-run never duplicates a menu.

SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL
Frameworks & APIs: React, Next.js, Tailwind CSS, Node.js, FastAPI, REST APIs, pytest, Puppeteer
Data & AI: PostgreSQL, Prisma, Supabase, pgvector, MinIO, Airflow, pandas, NumPy, Matplotlib, Gemini API, RAG
Cloud & Security: AWS (ECS, RDS, MWAA), Docker, Terraform, GitHub Actions, Git, Linux, MQTT, CVE/CVSS, SBOM